



Naturalistic food categories are driven by subjective estimates rather than objective measures of food qualities

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ABSTRACT

Food-related studies often categorize foods using criteria such as fat and sugar content (e.g., high-fat, high-sugar foods; low-fat, low-sugar foods), and use these categorizations for further analyses. While these criteria are relevant to nutritional health, it is unclear whether they agree with the ways in which we typically group foods. Do these objective categories correspond to our subjective sense? To address this question, we recruited a group of 487 online participants to perform a triplet comparison task involving implicit object similarity judgements on images of 36 foods, which varied in their levels of fat and sugar. We also acquired subjective ratings of other food properties from another set of 369 online participants. Data from the online triplet task was used to generate a similarity matrix of these 36 foods. Principal Components Analysis (PCA) of this matrix identified that the strongest determinant of food similarity (the first PC) was most highly related to participants' judgements of how processed the foods were, while the second component was most related to estimates of sugar and fat content. K-means clustering analysis revealed five emergent food groupings along these PC axes: sweets, fats, starches, fruits, and vegetables. Our results suggest that naturalistic categorizations of food are driven primarily by knowledge of the origin of foods (i.e., grown or manufactured), rather than by their sensory or macronutrient properties. These differences should be considered and explored when developing methods for scientific food studies.

1. Introduction

Research into food choice has been traditionally focused on the objective nutritional content of foods, especially levels of fat and sugar. These nutrients are both indicators of caloric density and thought to “incorporate the key variables believed to be responsible for influencing food intake, hunger, and food choice” (Blundell et al., 2009). Fat and sugar levels have been shown to be key drivers of both the metabolic and hedonic values of foods (DiFeliceantonio et al., 2018), and in turn, key drivers of the global obesity epidemic, which has been gaining popular attention since the early years of the 21st century. There is also evidence that fatness and sweetness are properties that affect the “palatability” judgements and activate dopamine reward pathways (Davis et al., 2008; Rada, Avena, & Hoebel, 2005).

As a consequence of this focus, food-related studies often group foods into separate categories based on relative fat and sugar content. For example, Finlayson et al (2007) sorted foods into four categories based on fat and sugar content in order to test whether ratings of “liking” and

“wanting” varied after eating a meal. Some widely used clinical assessments, such as the Leeds Food Preference Questionnaire (LFPQ) also broadly categorize foods based on fat and sugar content (Finlayson et al., 2007; Finlayson et al., 2008). However, while these nutrients reflect biologically relevant properties of foods, they are certainly not the only dimensions by which foods may vary. Moreover, they may not be the properties which are most relevant to individuals, or which influence their food choice.

The factors influencing food choice are broad, multifaceted, and frequently situational (Sobal & Bisogni, 2009). Included among a host of different factors are food-specific sensory properties, internal physiological states of the individuals, prior knowledge and attitudes about the food, and social and environmental factors external to the foods (Leng et al., 2017; Chen & Antonelli, 2020). Of the food-specific sensory properties, appearance, presentation, and texture are some of the major determinants of food preference and rejection in adults and children (Neumark-Sztainer et al., 1999; Sandvik et al., 2020; Laureati, 2022). Healthiness and nutrition claims are a strong cognitive factor driving

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food choice and even sensory enjoyment of foods (Provencher & Jacob, 2016; Oostenbach et al., 2019; Plasek et al., 2020), as is the perception that certain foods “go together” (Neumark-Sztainer et al., 1999). Importantly, the categorization of foods based on fat and sugar content can sometimes group foods which differ greatly in nutritional value, and which few would consider naturally “go together”. For instance, Finlayson (2007) grouped together marshmallows and fruit salad within their ‘low-fat sweet’ food category. Based strictly on the authors’ criteria, these foods are somewhat similar, in that they are both low in fat and contain sugar. However, apart from these properties, these foods could hardly be more different in terms of sensory properties like smell and texture as well as nutritional quality, not to mention other social and cultural qualities which might influence individual’s response to these foods.

A more naturalistic understanding of how people categorize food could help to build a more comprehensive picture of the relevant properties which underlie our food choices and, in turn, how the brain represents food-related information. One popular approach for obtaining naturalistic object groupings involves obtaining behavioral measures of the similarity of a group of objects. These measures can be implemented in overt tasks ranging from requiring participants to make pairwise ratings of object similarity based on specific criteria (Cortese & Dyre, 1996), or to form object groups by arranging them into clusters on a computer screen (Kriegeskorte & Mur, 2012; King et al., 2019), to more implicit measures such as requiring forced-choice comparisons of multiple items (Connolly et al., 2012; Hebart et al., 2020; Avery et al., 2022). The resulting pairwise similarity matrices from these tasks can be used, along with clustering and dimensionality reduction techniques, to identify emergent object categories or underlying dimensions on which those objects vary.

In the present study, we recruited an online sample of participants to judge the similarity of a set of food picture stimuli that contained equal groupings based on their sugar and fat content. Our central aims were to determine: 1) whether the natural categories generated from the similarity judgements reflect this categorization based on fat and sugar content, and 2) whether the underlying dimensions of these similarity judgements reflect the foods’ objective nutritional properties or more subjective assessments of those foods.

2. Methods

2.1. Food images

We selected a set of 36 food images from a larger set used in previous functional neuroimaging studies which examined the neural response to food images (Simmons et al., 2013, 2016; Avery et al., 2018). These foods were selected by the experimenters as examples of foods common

in a western American diet, foods that would be both familiar and instantly recognizable by participants with the United States. The foods selected by experimenters were divided into four macronutrient categories based on fat and sugar content (e.g., “High-Fat High-Sweet”, “High-Fat Low-Sweet”, “Low-Fat High-Sweet”, “Low-Fat Low-Sweet”; Fig. 1A). A variety of high-quality images of these foods were obtained through an online image search, which were then cropped and placed on a gray background. These images were subsequently normed by a separate pilot sample of 12 participants for naming accuracy, item-level typicality (as a representation of the specific food), category-level typicality (as a representation of the particular macronutrient category), as well as how appetizing the foods were (see Simmons et al., 2013 for details). Through this process, five images of 36 foods (9 per category) were selected that had the highest ratings of naming accuracy and typicality. For the present study, we selected one image per food that had the highest item-level typicality rating. The images and their associated sub-categories are shown in Fig. 1A (full versions of the images are available online, see Data/Code Availability statement).

2.2. Experiment 1: Food triplet task

We conducted an online behavioral study that investigated the salient semantic dimensions of our 36 foods using an odd-one-out triplet task (Avery, Liu, Carrington, & Martin, 2022; Connolly et al., 2012; Hebart, Zheng, Pereira, & Baker, 2020) on Amazon Mechanical Turk. Participants were shown a series of image triplets, consisting of three food images presented on screen that were randomly chosen from the set of 36 food pictures. Participants were instructed to click on the image that was the “odd one out.” For example, in one trial the participants might see pictures of apples, slices of bacon, and a carrot cake (Fig. 1B). One participant might choose the image of bacon as the odd one out, because cake and apples are both sweet foods. On the other hand, another participant might choose the apple image as the odd one out because the bacon and the cake images are more similar in color.

The triplet task has a variety of advantages, compared to other methods for assessing object similarity. Tasks involving explicit ratings of similarity along a particular dimension (e.g., fat and sugar) make assumptions about which properties are most important for judging object similarity. By remaining agnostic on what possible characteristics to evaluate the images on, the task allows us to analyze the implicit dimensions which underly participants’ similarity judgements in a data-driven manner. On the other hand, pairwise similarity tasks which don’t specify a dimension generally don’t account for how the similarity of objects varies with context, often in ways which are not immediately obvious (see Barsalou, 1982; Hebart et al., 2020 for more discussion). In this task, the similarity of any two foods will be judged in the context of all possible third foods to complete the triplet. For example, the pair of

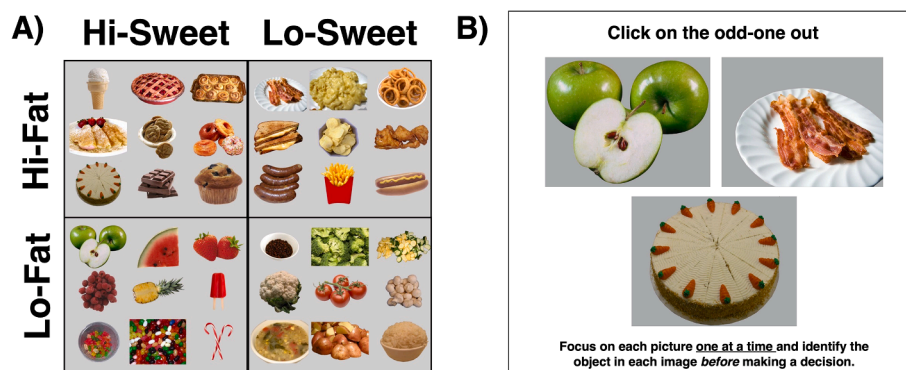


Fig. 1. Food Triplet Task. We presented online participants with a task in which they saw pictures of 36 foods that had been used in previous studies that examined the neural response to food images (Simmons et al., 2013, 2016; Avery et al., 2018) (A) These foods had been previously selected as examples of foods that were either high or low in fat and high or low in sugar. B) Those participants performed a triplet task where they selected the “odd-one-out” among multiple sets of three food images.

bacon and sausage will be judged in one triplet with soup, but also in other triplets with watermelon, and so on. By sampling across all possible triplets in this set of 36 foods (7,140 in total), we were able to sample across all the contexts and properties relevant to the foods in this dataset. Another advantage of this task is the reduced experimental burden on participants, compared to other tasks which require the sorting of large numbers of pictures into categories (Kriegeskorte & Mur, 2012; King et al., 2019). This factor, and the relative simplicity of the experimental presentation, make it ideal for online data collection in a large population of subjects.

Participants: A total of 487 online participants completed the triplet task (Age - mean (SD): 38.5 (12.0), range: 18–79; Gender: 214 Female, 17 no response). These participants all gave informed consent under the National Institutes of Health Institutional Review Board-approved protocol number 000589 and were compensated financially for their time. We restricted our participants to English speakers from within the continental United States. Additionally, we limited potential participants to Mechanical Turk users who had greater than 100 Human Interaction Tasks (HITs) approved and a HIT approval rate greater than 95 % for all requesters' HITs. In the context of mechanical Turk, a HIT is a virtual task that a requester releases and participants can accept, submit, and receive payment for completing. Each HIT contained 20 triplets.

One of the initial goals of this experiment was to verify the reliability of the triplet-derived similarity matrices across multiple sessions and multiple groups of individuals. The sample size for this experiment was thus the total number of individuals required to complete all HITs (max 714). We reviewed completed HITs in order to exclude non-human or inattentive participants. HITs were rejected if participants answered a question in under 250 ms on average or if a participant selected the photo in the same location for every triplet. We also added “catch” trials in each HIT. The catch trials were triplets that either consisted of two minus signs (“-”) and one plus sign (“+”), or two equals signs (“=”) and one plus sign. Participants were instructed to select the plus sign in these triplets. Each HIT contained one of each type of catch trial. HITs were rejected if participants failed a catch trial. Rejected HITs were republished so that they could be completed by another participant. Ultimately, we collected data from 629 participants over the course of two separate sessions and rejected 142 of those participants according to the above criteria, leaving 487 participants that completed a total of 13,580 triplets, with an average of 1.4 HITs (or 28 triplets) completed per participant. The order of triplets was randomized across participants, so each participant saw a unique series of triplets, allowing us to greatly minimize any order effects. Additionally, participants only completed a relatively small proportion of the total dataset (~0.2 % on average), allowing us to greatly minimize the potential effects of variability in food preferences.

2.3. Food similarity analysis

The responses to the triplet task were used to create a food similarity matrix, as follows (see Hebart et al., 2020; Avery et al., 2022). For each pair of foods, selected from our set of 36 foods, we calculated the similarity of those foods as the proportion of triplets containing those foods in which neither food was selected as the “odd-one-out.” For example, to calculate the similarity between apples and bacon, we first found the number of triplets that included both apples and bacon. We then determined the subset of those triplets in which neither apples nor bacon were selected as the “odd one out.” Since neither apples nor bacon were selected, we can conclude that these foods were judged to be more similar to each other than to the third food that was presented. We divided the subset in which neither apples nor bacon were picked by the total number of triplets that included apples and bacon. The resulting number was between 0 and 1. A number closer to 1 indicates a higher degree of similarity between the two foods. The result of this process was a 36 by 36 matrix reflecting the similarity of all food pairs.

We created the similarity matrices for each of the triplet datasets using the statistical software package R (<https://www.r-project.org>) using custom code and functions (see Data Availability Statement). We initially created separate similarity matrices for our two datasets in order to compare the matrices across sessions. To do so, we calculated the Pearson correlation of the upper-triangular portions of the two similarity matrices. We then averaged those two matrices for the analyses which follow.

2.4. Dimensionality reduction and clustering analyses

We used a series of dimensionality reduction and clustering techniques to identify the most relevant underlying dimensions of our food similarity matrix as well as the natural food categories that emerge from these behavioral judgements of food similarity. Firstly, we ran Principal Components Analysis on the food similarity matrix, using the *prcomp* function in the R stats package. Using this function, we identified the top dimensions which accounted for the greatest amount of variance in these data (~80 %) and used those for the majority of our further analyses. For our clustering analyses, we used the silhouette method, implemented in the *fviz_nbclust* function from the *factoextra* R package to identify the optimal number of clusters to use for k-means clustering, which we applied to the Principal Components data, using the R stats function *kmeans*. The silhouette method involves the calculation of the average silhouette coefficient, a measure of within vs. between cluster similarity, for a range of possible cluster values. The optimal cluster value is thus the value with largest average silhouette coefficient (Rousseeuw, 1987).

2.5. Experiment 2: Food property ratings

Following this, we sought to identify what food-relevant dimensions might be represented within the food similarity space generated from our online behavioral data. We observed that the clustering of the foods along the Principal Component (PC) dimensions produced a distinct set of groupings from the original Fat-Sugar groupings. While the second PC did seem to separate sweet foods from non-sweet foods, the first Principal Component did not seem to separate foods based on either fat or sugar content, but by some qualitative measure such as how processed or healthy those foods are. To investigate this further, we collected data on both subjective and objective properties of these foods, theorizing that the PC dimensions would show a greater relationship to subjective estimates of food properties than to objective measures of nutritional content.

We collected another set of online behavioral data about perceived levels of processing (i.e., how processed is this food?), healthiness, fat content and sugar content for all of the 36 foods on Mechanical Turk. For each of these metrics, we collected a separate sample of 100 online participants. This sample size was determined, using the *pwr* power calculation library in R (<https://cran.r-project.org/web/packages/pwr/pwr.pdf>), to provide a power of ~ 1 (i.e., type II error probability = 1×10^{-9}) for a one-sample *t*-test with a large effect size (Cohen's *d* = 0.8) and an alpha = 0.05. Participants rated each food using a Likert scale which ranged from 1 “Not processed/healthy at all” or “No sugar/fat” to 7 “Very processed/healthy” or “Extremely high in sugar/fat”. We used the same participant exclusion criteria as above for the triplet task. Each participant rated each of the 36 foods in each HIT, along with 4 catch trials, where subjects were instructed to select a specific number on the Likert scale. The HITs were reviewed with the same criteria as the triplet task, meaning that HITs rejected for failing catch trials, pressing the same button, or too short reaction times (< 0.5 s). Additionally, we obtained data on typical portion size as well as calorie, sugar, fat, saturated fat, and protein content per 100 g for all of the 36 foods from US Department of Agriculture's Food Data Central (USDA, 2019; <https://fdc.nal.usda.gov/index.html>).

Participants: A total of 369 online participants completed these

rating tasks (Female: 131, no response: 1). Participants were required to be high school graduates, native English speakers, and live within the United States. These participants gave informed consent under the National Institutes of Health Institutional Review Board approved protocol number 000589 and were compensated financially for their time.

Using the online ratings, we calculated average subjective estimates of the fat content, sugar content, healthiness, and degree of processing associated with each of our foods. We compared the subjective ratings of food properties and objective measures of nutrient composition to the Principal Components of our food similarity matrix. For both Principal Components, we created separate subjective and objective linear models using the R stats *glm* function and identified which model best fit the data by comparing the adjusted r-squared and Akaike Information Criteria (AIC) values of the respective models and using Wilks' likelihood ratio test to compare their goodness of fit, implemented in the function *lrttest* from the *lmtest* R package. Following this step, we also identified which objective and subjective variables were most highly correlated with the PCs, by calculating Pearson correlations for each of those variables and correcting the resulting p-values for multiple comparisons using a False-Discovery-Rate correction (Benjamini & Hochberg, 1995).

2.6. Relationship between objective and subjective ratings

Finally, we sought to examine the relationship between the objective fat and sugar content of our 36 foods and subjective ratings of those properties. Specifically, we examined whether the relationship between nutrient content (in g/100 g) and estimates of nutrient content was best characterized as linear or logarithmic. To do so, we used Wilks' likelihood ratio test to compare the goodness of fit of linear and logarithmic models of the average fat and average sugar ratings from our food rating data. To enable log transformation, we added 1 to the distributions of fat and sugar content, as some of the 36 foods had 0 g of fat or sugar per serving. We also compared the average adjusted r-squared values of linear and logarithmic models across the groups of 100 subjects who provided fat or sugar ratings using a paired *t*-test.

3. Results

3.1. Food triplet task

We collected two datasets of the triplet task over two separate data collection sessions. Dataset 1 contained a total of 6460 and Dataset 2

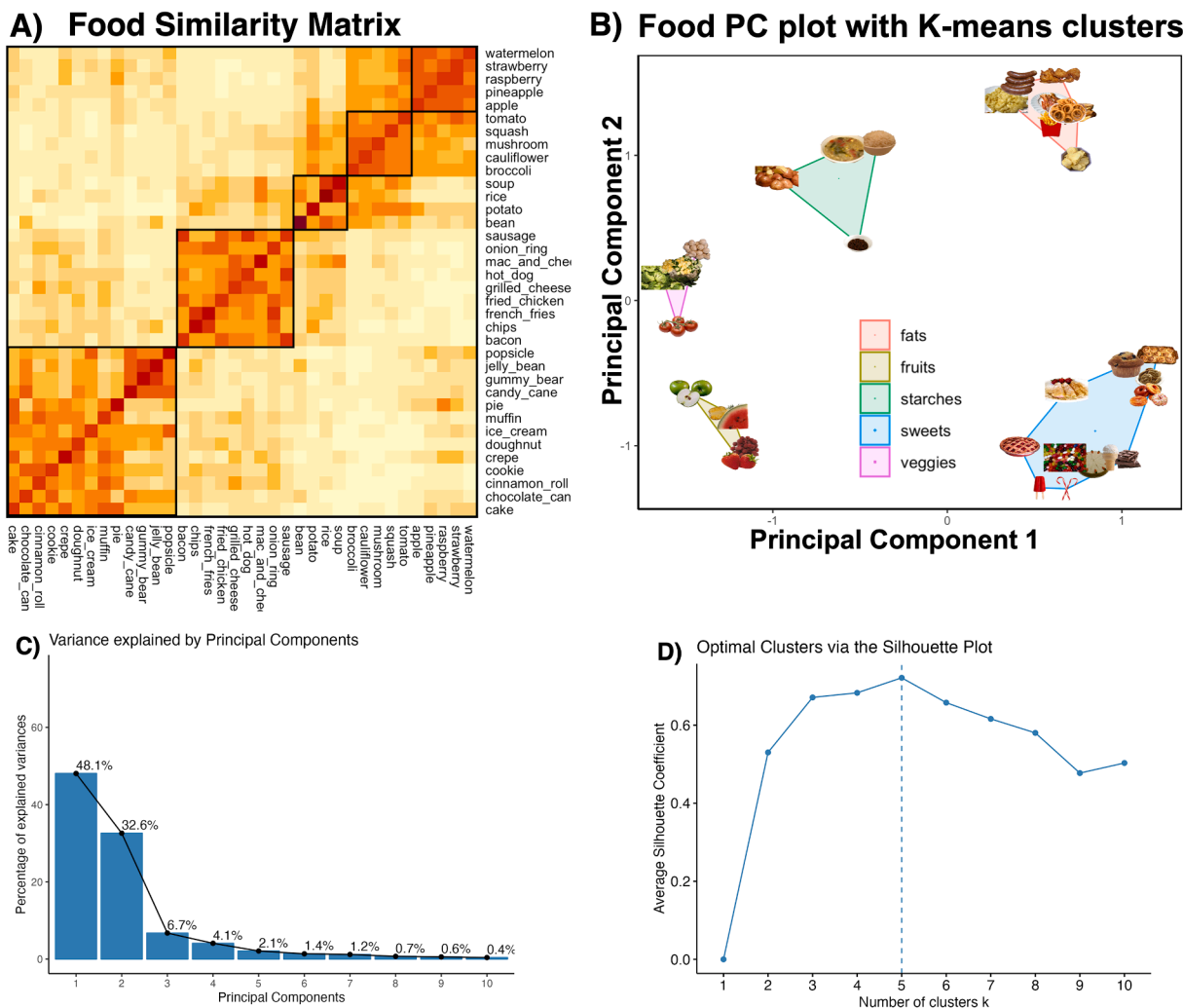


Fig. 2. Food Similarity Clustering. We used the triplet task data to calculate the similarity between every possible combination of our 36 foods. The similarity between any two foods was calculated as the proportion of triplets that included the two foods in which the third food in the triplet was chosen as the "odd-one-out." (A) The similarity matrix is represented by a heat map in which darker, redder colors represent a higher degree of similarity. (B) We performed Principal Components Analysis (PCA) on the triplet data and plotted the first two PCs against each other, before performing k-means clustering to identify naturalistic food categories. The five emergent food categories are depicted using colorful shading and are also outlined in black in the similarity matrix. (C) A plot of variance explained by the PCs. (D) A silhouette plot used to define the optimal number of clusters.

contained 7120 triplets, for a total of 13,580 triplets, after removal of HITs per our exclusion criteria. A total of 6440 of those triplets were repeated across sessions. We created 36x36 food similarity matrices for both of these datasets. For this task, we calculated the similarity between any two foods as the percentage of triplets that included the two foods in which the third food was selected as the “odd one out.” We identified that the correlation of the upper triangular portion of these matrices was extremely high ($r = 0.91$, $p < 0.001$). As a result, we combined these two matrices for the analyses which follow (Fig. 2a).

3.2. Clustering analyses

We performed PCA on the combined food similarity matrix and identified that the first two Principal Components accounted for over 80 % of the variance in the data (Fig. 2). We performed our clustering analyses on these first 2 PC dimensions, for ease of visualization. We identified, via the silhouette method, that the optimal number of clusters for these data was 5 (Fig. 2) and used that result when applying k-means clustering to the Principal Component data. This process thus clustered together foods according to easily recognizable food categories, such that all vegetables were clustered together, as were all fruits (though tomatoes were grouped with vegetables, not fruits). High fat, savory foods like French fries, sausages, and fried chicken were clustered together as were all high-fat sweet foods. Notably, low-fat sweet foods like candy canes and jelly beans were instead grouped along with the high-fat sweet foods, and not with the fruits. Low-fat starches like potatoes, rice, and beans formed a separate cluster. We identified the five categories which emerged from this process as Sweets, Fats, Starches, Fruits, and Vegetables (Fig. 2).

3.3. Food property ratings

We collected ratings of perceived fat content, sugar content, healthiness, and level of processing (i.e., “how processed is this food”) associated with each of our 36 foods from separate groups of online participants (369 total). The average of these ratings for each food are listed in Table 1.

Of note, ratings of healthiness and levels of processing were almost perfectly negatively correlated with one another ($r = -0.98$, $p < 0.001$). We generated separate ‘subjective’ and ‘objective’ models to account for variance in the first 2 Principal Components from the food similarity matrix (see Methods). The objective model listed the protein, sugar, fat, carbohydrate, and total caloric content per 100 g serving of each food (PC1: adjusted $r^2 = 0.78$, AIC = 64.9; PC2: adjusted $r^2 = 0.67$, AIC = 60.7). The subjective model contained the average ratings of the four food property estimates we collected (PC1: adjusted $r^2 = 0.97$, AIC = -6.9; PC2: adjusted $r^2 = 0.90$, AIC = 16.1). The Wilk’s likelihood ratio test identified that the fit of the subjective model was significantly greater than the objective model for both PC1 ($X^2 = 69.8$, $p < 0.001$) and PC2 ($X^2 = 42.6$, $p < 0.001$).

The highest correlation for PC1 was with estimates of how processed ($r = 0.97$, $p < 0.001$; Fig. 3A) and how healthy the foods were ($r = -0.96$, $p < 0.001$; Fig. 3B), followed by estimated fat ($r = 0.88$, $p < 0.001$) and calories/100 g ($r = 0.88$, $p < 0.001$). PC2 was most highly correlated with estimated sugar content ($r = -0.78$, $p < 0.001$; Fig. 3C) and sugar/100 g ($r = -0.63$, $p < 0.001$). Though, the high fit of the subjective model fit for PC2 was determined by both estimated sugar ($\beta = -0.59$, $p < 0.001$) and estimated fat ($\beta = 0.34$, $p < 0.001$; Fig. 3D). See Table 2 for a full list of correlations.

Table 1
Nutrient data, categorization, and subjective ratings of foods used in online behavioral experiment.

Foods	Cluster	Category	USDA Data					Subjective Ratings Data			
			kcal/100 g	fat/100 g	carbs/100 g	sugar/100 g	protein/100 g	Fat	Sugar	Processed	Health
Apple	fruits	LFHS	52.0	0.17	13.81	10.39	0.26	2.29	4.28	2.88	6.24
Bacon	fats	HFLS	468.0	35.10	1.70	0.00	33.90	5.77	2.96	4.91	3.40
Bean	starches	LFLS	46.0	0.66	7.71	1.24	2.42	3.20	3.22	3.54	5.31
Broccoli	veggies	LFLS	35.0	0.41	7.18	1.39	2.38	2.39	2.43	2.92	6.39
Cake	sweets	HFHS	429.0	26.71	45.74	34.04	4.07	5.53	6.36	5.49	2.98
Candy cane	sweets	LFHS	357.0	0.00	100.00	78.57	0.00	3.49	5.69	5.74	2.28
Cauliflower	veggies	LFLS	25.0	0.28	4.97	1.91	1.92	2.43	2.25	2.86	6.29
Chips	fats	HFLS	532.0	33.98	53.83	0.33	6.39	5.33	3.04	5.63	2.57
Chocolate candy	sweets	HFHS	535.0	29.66	59.40	51.50	7.65	5.03	6.29	5.44	3.17
Cinnamon roll	sweets	HFHS	452.0	26.61	48.60	25.70	4.45	5.57	6.08	5.68	2.75
Cookie	sweets	HFHS	473.0	23.82	58.15	28.58	8.92	5.03	5.50	5.39	2.99
Crepe	sweets	HFHS	183.0	6.95	26.38	16.25	4.56	5.40	5.65	5.31	3.51
Doughnut	sweets	HFHS	421.0	22.70	47.93	22.75	6.14	5.70	6.34	5.98	2.42
French fries	fats	HFLS	289.0	14.04	37.20	0.28	3.49	5.75	3.06	5.75	2.52
Fried chicken	fats	HFLS	289.0	17.35	9.42	0.00	22.50	6.04	3.00	5.19	3.17
Grilled cheese	fats	HFLS	331.0	19.22	28.70	5.03	10.80	5.37	4.14	5.34	3.32
Gummy bear	sweets	LFHS	300.0	0.00	72.00	48.00	5.00	3.81	5.90	5.87	2.30
Hot dog	fats	HFLS	308.0	18.20	23.81	3.95	11.20	5.55	3.91	5.90	3.00
Ice cream	sweets	HFHS	207.0	11.00	23.60	21.22	3.50	4.71	6.13	5.54	2.69
Jelly bean	sweets	LFHS	357.0	0.00	89.29	67.86	0.00	3.69	6.14	5.80	2.44
Mac and cheese	fats	HFLS	164.0	4.99	23.10	1.57	6.68	5.51	3.83	5.56	2.95
Muffin	sweets	HFHS	375.0	16.07	53.00	31.47	4.49	5.04	5.83	5.43	3.24
Mushroom	veggies	LFLS	22.0	0.34	3.26	1.98	3.09	2.67	2.51	2.91	6.20
Onion ring	fats	HFLS	411.0	25.23	43.58	5.43	3.86	5.45	3.13	5.53	2.66
Pie	sweets	HFHS	260.0	11.00	39.80	14.29	2.00	4.54	5.88	4.99	3.84
Pineapple	fruits	LFHS	50.0	0.12	13.12	9.85	0.54	2.48	4.90	2.95	6.15
Popsicle	sweets	LFHS	81.0	0.24	19.68	14.57	0.00	3.48	5.83	5.38	2.40
Potato	starches	LFLS	87.0	0.15	19.59	1.43	2.30	3.06	3.21	3.07	5.42
Raspberry	fruits	LFHS	52.0	0.65	11.94	4.42	1.20	2.88	4.72	2.88	5.80
Rice	starches	LFLS	124.0	0.50	26.76	0.00	2.18	3.31	3.07	3.91	5.04
Sausage	fats	HFLS	325.0	27.25	1.42	1.09	18.50	5.68	3.56	5.25	3.41
Soup	starches	LFLS	65.0	0.90	12.11	4.31	2.20	3.32	2.70	4.21	5.72
Squash	veggies	LFLS	17.0	0.32	3.11	2.50	1.21	2.38	2.88	2.96	6.31
Strawberry	fruits	LFHS	32.0	0.30	7.68	4.89	0.67	2.45	4.66	2.87	6.16
Tomato	veggies	LFLS	18.0	0.20	3.89	2.63	0.88	2.44	3.11	3.00	6.28
Watermelon	fruits	LFHS	30.0	0.15	7.55	6.20	0.61	2.19	4.62	2.71	6.03

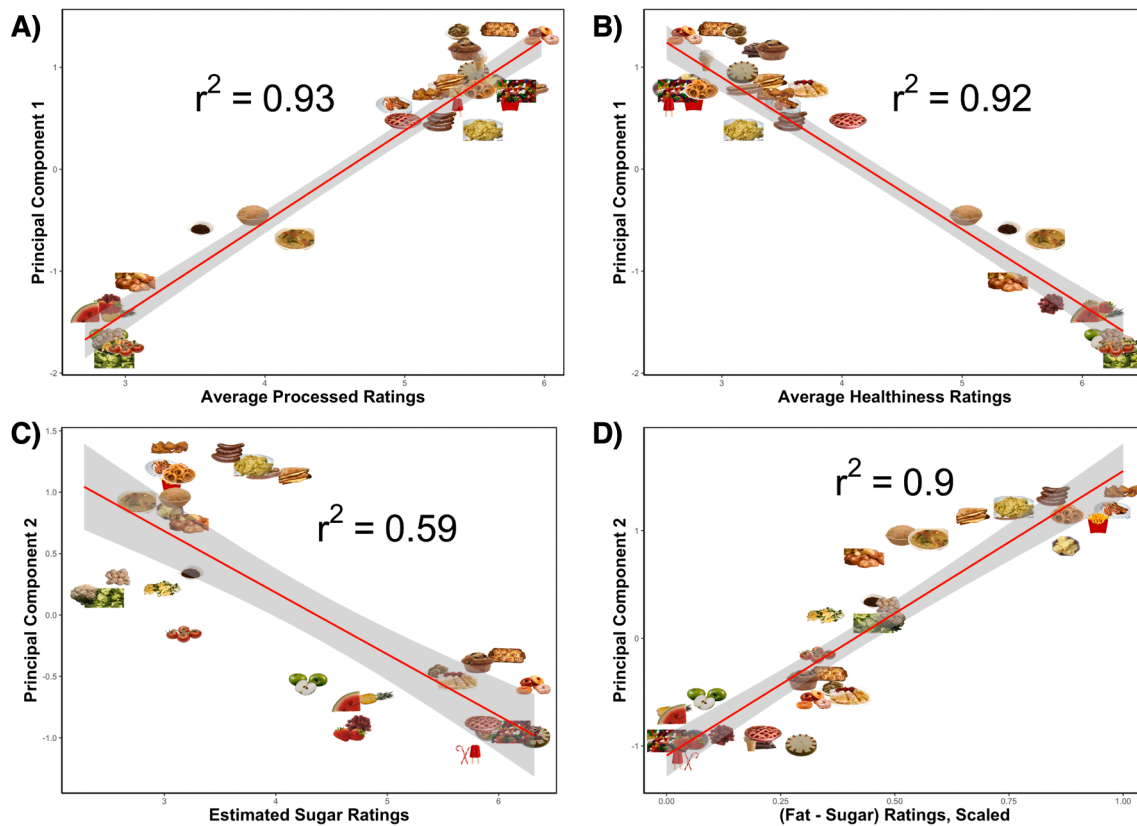


Fig. 3. Food Property Ratings. We compared the subjective ratings we gathered from our online participants to the Principal Components generated from the food triplet data. (A) The first Principal Component from the triplet data was nearly perfectly correlated with participants' estimates of how "processed" each food was, (B) and negatively correlated with participants' estimates of how "healthy" each food was. The "processed" and "healthiness" ratings were themselves highly negatively correlated ($r = -0.98$). (C) The second Principal Component from the triplet data had the highest relationship to ratings of estimated sugar content, (D) though a combined model incorporating both estimated fat and sugar ratings accounted for almost all of the variance in this Principal Component.

Table 2

Correlations between Principal Components and Nutritional and Rating Data.

Values	PC1			PC2		
	r	p	pFDR	r	p	pFDR
Calories/100 g	0.88	<0.001	<0.001	0.06	0.73	0.73
Fat/100 g	0.70	<0.001	<0.001	0.29	0.09	0.09
Carbs/100 g	0.65	<0.001	<0.001	-0.38	0.02	0.02
Sugar/100 g	0.47	0.004	0.005	-0.63	<0.001	<0.001
Protein/100 g	0.39	0.019	0.021	0.53	<0.001	<0.001
Processed Rating	0.97	<0.001	<0.001	0.05	0.76	0.76
Fat Rating	0.88	<0.001	<0.001	0.35	0.04	0.04
Sugar Rating	0.56	<0.001	<0.001	-0.78	<0.001	<0.001
Health Rating	-0.96	<0.001	<0.001	0.00	0.99	0.99
Appetizing	0.45	0.006	0.007	-0.2	0.24	0.40

3.4. Relationship between objective and subjective ratings

Finally, we tested whether the relationship between objective nutrient content and subjective estimates of that content for our 36 foods was linear or logarithmic. We identified that the logarithmic model was a significantly better fit between average ratings of fat and grams of fat/100 g (linear: adjusted $r^2 = 0.69$, AIC = 83.2; log: adjusted $r^2 = 0.84$, AIC = 59.7; Wilk's likelihood ratio: $X^2 = 23.5$, $p < 0.001$; Fig. 4A) and between average ratings of sugar and grams of sugar/100 g (linear: adjusted $r^2 = 0.53$, AIC = 101.3; log: adjusted $r^2 = 0.79$, AIC = 72.5; Wilk's likelihood ratio: $X^2 = 28.8$, $p < 0.001$; Fig. 4C). We also verified this result across individuals as well for fat (linear: avg. $r^2 = 0.38$; log: avg. $r^2 = 0.46$; $t = 13.9$, $p < 0.001$; Fig. 4B) and sugar ratings (linear: avg. $r^2 = 0.30$; log: avg. $r^2 = 0.43$; $t = 13.5$, $p < 0.001$; Fig. 4D).

3.5. Supplementary analyses

We performed a variety of supplementary analyses to account for alternate interpretations of our results and potential confounds. Firstly, we sought remove the potential confound of fat content from our original analysis by separately regressing out actual fat content and estimated fat content from processed ratings. We then examined the relationship between PC1 and those separate sets of residuals. After regressing out actual fat content (P_A) and estimated fat content (P_E), the ratings of perceived processing were still significantly related to PC1 (P_A : $r = 0.62$, $p < 0.001$, adjusted $r^2 = 0.45$; P_E : $r = 0.42$; $p = 0.01$, adjusted $r^2 = 0.15$). Thus, even after accounting for the fat content of the foods, perceived level of processing accounted for as much as 45 % of the variance of PC1.

We also examined the relationship between healthiness ratings and processed ratings, after regressing out the fat content of the foods (both actual and estimated). Accordingly, after regressing out actual fat content (P_A) and estimated fat content (P_E), the processed ratings were still significantly negatively related to healthiness ratings (P_A : $r = -0.77$, $p < 0.001$, adjusted $r^2 = 0.57$; P_E : $r = -0.53$; $p < 0.001$, adjusted $r^2 = 0.26$).

Finally, we examined the relationship between PC1 and PC2 and ratings of how appetizing the foods in those pictures were, provided by a normative sample of subjects in a previous study (Simmons et al., 2013). Appetizing ratings were significantly correlated to PC1, but not PC2 (PC1: $r = 0.45$, $p = 0.006$; PC2: $r = -0.2$, $p = 0.24$). However, we found that including appetizing ratings in a subjective model of PC1 (along with processed, health, fat, & sugar ratings) did not explain any additional variance in PC1 (ANOVA: $F = 1.53$, $p = 0.23$).

We additionally examined whether ratings of the level of processing of these foods were consistent with the NOVA categorization of these

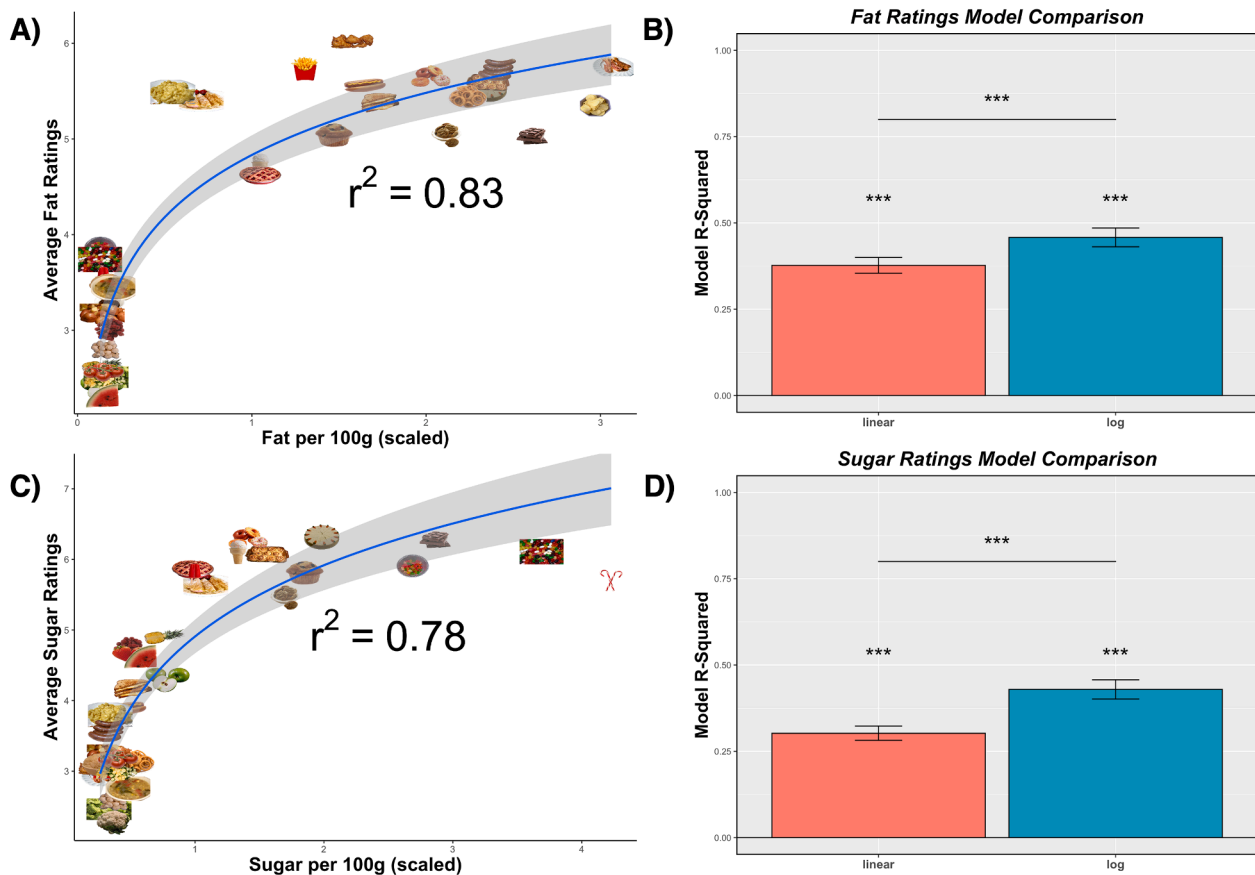


Fig. 4. Log-Linear relationship between objective and subjective ratings. We further analyzed the relationship between the subjective food property ratings and the objective nutrient data of these foods, obtained from a USDA database. (A) We identified a logarithmic relationship between the average estimated fat ratings of these foods and their actual fat content per 100 g serving. (B) We observed a similar relationship between the average sugar ratings and the actual sugar content per 100 g serving. (C) Across online participants, the relationship between fat and sugar ratings and their objective counterparts were better explained using a logarithmic model, rather than a linear model.

foods, which we obtained via an online database (<https://world.ope.nfoodfacts.org/nova>). Our participants rated the “unprocessed” foods as significantly less processed than foods categorized as either “processed” or “ultra-processed”, ($t = 16.9$, $p < 0.001$), suggesting they are able to reliably identify the level of processing involved in the preparation of these foods.

4. Discussion

We set out to test whether the way individuals categorize foods adheres to traditional groupings based on fat and sugar content. To explore this question, we recruited online participants to judge the similarity of a set of food images that varied in their fat and sugar content, using an implicit similarity judgement task. In another experiment, we also collected sets of online ratings of specific properties of these foods. The analysis of these similarity judgments allowed us to determine how individuals would naturally categorize these foods, what specific dimensions underlie these food categories, and the relationship between objective and subjective estimates of those foods’ fat and sugar content.

The similarity judgements that we collected on our 36 foods revealed naturalistic categories that differ from those based purely on fat and sugar content. The original “low-fat high-sweet” sub-category included fruits such as strawberries and watermelon as well as candies like gummy bears and jellybeans. Within the similarity space generated though analysis of our triplet task data, we found that this group had split. Fruit emerged as a sub-category on its own, while processed sweets like gummy bears, jellybeans, and candy canes were absorbed into a

grouping that contained “high-sweet high-fat” foods like pies, cakes, and doughnuts. Additionally, the “low-fat low-sweet” group broke off into two separate groups, vegetables (mushroom, cauliflower, squash, tomato) and starches (rice, beans, soup, potatoes). These naturalistic categories are thus much more reminiscent of the classical “food groups”, such as those included in the USDA guidelines as a basis for healthy diet (for reference, see myplate.gov).

We used Principal Component Analysis on the food similarity matrix and found that the first and second components accounted for over 80 % of the overall variance. In order to identify what food properties were represented by these components, we generated an “objective” model composed only of the macronutrient values of the foods from a USDA database and a “subjective” model composed of estimated ratings of those values from an independent online participant sample, along with estimates of the healthiness and level of processing of those foods. We identified firstly that the “subjective” model provided a significantly better fit to both Principal Components, as might be expected, given the subjective nature of the food similarity judgements. We then identified which specific factors of the “subjective” model best predicted each component. The first component was nearly perfectly correlated to qualitative ratings of how processed and (un)healthy those foods were. This result is supported by another recent study that examined food image similarity in an online sample, which reported that food similarity was most highly correlated with ratings of the naturalness and the perceived transformation (i.e., processing) of food images (Moerel et al., 2023). The importance of this distinction between “processed” and “unprocessed” foods may be related to findings in studies of object property knowledge that emphasized “artificial” and “organic” as among the top

three most salient object property dimensions (with “food-related” as number two; [Hebart et al., 2020](#)). The second Principal Component was driven by subjective estimates of both sugar and fat content. When plotting foods along the first two principal components of their similarity matrix, the distinction between the original and naturalistic food categories becomes clear. The original sweet-fat dimensions become squashed along a single axis, and what primarily distinguishes between those foods is their perceived healthiness or artificiality.

Interestingly, healthiness ratings of the foods and “processed” ratings of the foods were themselves almost perfectly anti-correlated, which suggests how tightly interconnected those concepts may be in the minds of the general public. Notably, processed ratings and healthiness ratings were not simply a proxy for fat content, as while the majority of the foods rated as highly processed (rating 5 or greater) were also high in fat, this was not uniformly true, other foods (popsicle, candy cane, jellybeans, and gummy bears) which were high in sugar and contained no fat were also rated as highly processed. Indeed, the tight relationship between PC1, health ratings, and processed ratings remained even after regressing out estimated and actual fat content of the foods. This result suggests that individuals have a general impression that less processed, or more ‘natural’ foods are healthier, independent of fat or sugar content. While this impression seems to reflect the beliefs of our sample population, it might well vary across cultures, or across individuals with different nutritional needs. For example, for underweight individuals, processed foods may be a healthier and easier to consume source of calories than raw, unprocessed foods. Future studies examining these relationships might be able to address this point more comprehensively.

Our analysis of the relationship between the objective nutrient content and subjective ratings yielded further insights into how individuals infer food properties. The subjective fat and sugar ratings we collected from our online participants exhibited a logarithmic relationship to the objective fat and sugar content we sourced from the USDA, rather than a direct linear relationship. This observation suggests that these food property inferences follow a transformation from their objective counterparts similar to Fechner’s law (often referred to as the Weber-Fechner law, but see [Algom, 2021](#)) of perceptual psychophysics, which states (roughly) that the perceived intensity of a stimulus property increases as the logarithm of the increase of that property, rather than linearly ([Fechner, Howes, & Boring, 1966](#); [Dehaene, 2003](#)). This relationship has been demonstrated in perceptual ratings of foods such as the self-reported sweetness intensity of sweet tastants ([Mao et al., 2019](#)) and the perceived thickness of liquid foods ([Deblais et al., 2021](#)). This connection suggests that, when individuals view pictures of foods and infer food properties, their responses closely follow their direct perceptual counterparts. Importantly, these results suggest that individuals’ estimates become increasingly insensitive to increases in fat and sugar with higher concentrations. Thus, while they may accurately discern between foods ‘low’ and ‘high’ in fat (e.g., baked potatoes and French fries), they are much less accurate between foods ‘high’ and ‘very high’ in fat (e.g., French fries and potato chips). This relative insensitivity, on the order of ~ 200–300 calories, could easily lead to dramatic increases in caloric intake and weight gain over time.

While the approach we used for the present study was designed to address our particular research question, it does come with some limitations. First, our sample of foods was limited to pictures of only 36 foods. This allows us to only sample a relatively small portion of food similarity space, so it may not give us the broadest picture of food conceptual knowledge, nor the full space of relevant food categories, for at least this segment of the population. A wider variety of foods would perhaps have yielded new or different emergent dimensions that, for one reason or another, were not as salient in discriminating among the 36 foods that we used. For instance, with more meat items in the set, we might expect a ‘meat’ category to separate out from the rest of the foods. Further study and validation using larger samples of natural foods, such as those present in the THINGS dataset ([Hebart et al., 2020](#)) or in other large sets of publicly available food stimuli ([Blechert et al., 2019](#)), could

help validate and expand upon the results of this study. The selected foods were also somewhat specific to a western American diet, and the groupings of foods might vary quite a bit across cultures with different cuisines and dietary practices. Also, while we limited our pool of online participants to English-speakers within the continental United States, intra-regional cultural differences within the US could still influence participants’ perspective on the food items we used. Further study across cultures using similar techniques might help to identify which food categories and which underlying dimensions are shared or distinct across cultures. Ultimately, these particular foods were chosen as an example of a previously generated set of stimuli used for a food-related fMRI study, defined according to a specific organizational principle. Our methods and analyses allowed us to test whether that organizational principle was a reasonable method of sampling the relevant properties underlying conceptual knowledge about food.

5. Conclusion

The results of the present study suggest that categorization of foods based exclusively on fat and sugar content does not correspond well to how individuals naturally categorize food. Our results suggest that, while subjective estimates of fat and sugar do relate to how we discriminate foods, individuals prioritize properties such as healthiness and artificiality of foods above those foods objective nutrient composition. The most immediate question that could be addressed by these findings would be whether the neural response to food pictures corresponds more to these naturalistic categories or to categories based on fat and sugar content. Through the use of information-mapping techniques such as Representational Similarity Analysis ([Kriegeskorte et al., 2008](#)) or voxel-wise encoding models ([Naselaris et al., 2011](#)), previous studies have mapped behavioral similarity judgements onto the neural structures involved in representing their information ([Connolly et al., 2012](#); [King et al., 2019](#)). Identifying where and how food-relevant properties are represented will greatly aid our understanding of how food-related information is distributed in the brain, and in turn how dysfunction within specific regions of that distributed network may underlie the pathophysiology of eating-related mental health disorders.

6. Ethics statement

All participants of our study gave informed consent under the National Institutes of Health Institutional Review Board-approved protocol number 000589 and were compensated financially for their time.

7. Data/code availability

The raw datasets, analysis code, and image sets for the current study have been placed in a public repository (<https://doi.org/10.17605/OSF.IO/8DS7G>).

CRediT authorship contribution statement

Madeline Carrington: Data curation, Investigation, Writing – original draft. **Alexander G. Liu:** Software, Investigation, Validation. **Caroline Candy:** Data curation, Investigation, Validation. **Alex Martin:** Supervision, Funding acquisition, Writing – review & editing. **Jason Avery:** Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The raw datasets and analysis code for the current study have been deposited in a public repository (<https://doi.org/10.17605/OSF.IO/8DS7G>).

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